

Alexis Sáez

Postdoctoral Scholar
California Institute of Technology (Caltech)
Division of Geological and Planetary Sciences
1200 E California Boulevard, Pasadena, CA, USA 91125
Phone: (+1) 626 989 2182, E-mail: saez@caltech.edu

EDUCATION

EPFL - École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

Ph.D. in Mechanics

2019-2023

- Advisor: Prof. Brice Lecampion.
- Research areas: Geomechanics, induced seismicity, earthquake and fracture mechanics, theoretical and computational mechanics.
- Awarded the Rocha Medal 2026 by the International Society for Rock Mechanics and Rock Engineering (ISRM) as the best PhD thesis in the field, worldwide, defended in 2023-2024.
- Awarded as the best PhD thesis in Mechanics in EPFL in 2024 (EDME award).

University of Chile

Santiago, Chile

M.Sc.E., B.Sc.E., and professional title in Civil Engineering

2005-2014

- Awarded as the Best Graduate Civil Engineer of the 2014 Class.

PROFESSIONAL EXPERIENCE

California Institute of Technology | Division of Geological and Planetary Sciences

CA, USA

Postdoctoral Scholar Research Associate in Geology

2024-Today

- Advisor: Prof. Jean-Philippe Avouac. Co-advisor: Prof. Ares Rosakis.
- Collaborators: Prof. Nadia Lapusta.
- Research areas: Geophysics, earthquake source physics, slow and fast earthquakes, experimental fault mechanics, induced seismicity, geomechanics for geo-energy applications.
- Awarded the Swiss NSF Postdoc.Mobility Fellowship.
- Awarded the Caltech Geophysics Option Postdoctoral Fellowship.

Dalhousie University | Department of Civil and Resource Engineering

NS, Canada

Postdoctoral Fellow in Civil and Resource Engineering

2024

- Advisor: Prof. Dmitry Garagash.
- Research areas: Earthquake source physics, slow earthquakes, theoretical mechanics.

EPFL - École Polytechnique Fédérale de Lausanne | Institute of Civil Engineering

Lausanne, Switzerland

Doctoral Researcher in Mechanics

2019-2023

- Advisor: Prof. Brice Lecampion.
- Research areas: Geomechanics, induced seismicity, earthquake and fracture mechanics, theoretical and computational mechanics.
- Awarded the Swiss Government Excellence Scholarship for Foreign Students.

The University of Tokyo | Earthquake Research Institute

Tokyo, Japan

Visiting Doctoral Researcher in Geophysics

2022-2023

- Advisor: Prof. Aitaro Kato.
- Research areas: Geophysics, slow earthquakes.

- Awarded the EPFL Doc.Mobility Grant.

CODELCO - Corporación Nacional del Cobre de Chile

Santiago, Chile

Expert Consultant

2025-Today

- Consultancy on mining-induced seismicity and hydraulic fracturing operations.

Project Engineer

2013-2016

- Structural, geomechanical, and seismic engineering revisions for large mining projects and tailings dams.
- Management of engineering mining projects, techno-economic analysis, and strategic planning.

AWARDS

2025	Rocha Medal 2026: given by the International Society for Rock Mechanics and Rock Engineering (ISRM), to the best PhD thesis in Rock Mechanics and Rock Engineering worldwide, defended in 2023-2024.
2024	2024 EDME award: given by the Doctoral Program in Mechanics (which includes biomechanics, geomechanics, fluid mechanics, and solid mechanics) of the École Polytechnique Fédérale de Lausanne, to the best PhD thesis of the 2024 class.
2014	Best Graduate Engineer Award: given by the Chilean Society of Engineers (Colegio de Ingenieros de Chile) to the best graduate civil engineer of the 2014 class of the University of Chile.
2006, 2007, 2008, 2009, 2010	Outstanding Student Award: granted each year by the Faculty of Physical and Mathematical Sciences of the University of Chile, to undergraduate students belonging to the 6% of higher grades of the faculty.

GRANTS AND SCHOLARSHIPS

2026-2027	GMG Grant: granted by the Industrial Advisory Board of the Center for Geomechanics and Mitigation of Geohazards (GMG) at Caltech, to conduct research on the development of theoretical models of injection-induced fault slip. Amount: ~ 40,000 USD. Role: Principal Investigator (PI).
2025-2027	SNSF Postdoc Mobility Fellowship: granted by the Swiss National Science Foundation (SNSF) to conduct research at Caltech in geology and geophysics. Amount: ~ 155,000 USD.
2024-2025	Caltech Geophysics Option Postdoctoral Fellowship: granted by the Caltech Division of Geological and Planetary Sciences to work on the broad field of geophysics. Amount: ~ 70,000 USD.
2022-2023	EPFL Doc Mobility Grant: given by the École Polytechnique Fédérale de Lausanne (EPFL) to carry out a doctoral research stay of 6 months in the Earthquake Research Institute (ERI) of the University of Tokyo. Amount: ~ 10,000 USD.
2019-2023	Swiss Government Excellence Scholarship for Foreign Students: granted by the Swiss Confederation to conduct research at the École Polytechnique Fédérale de Lausanne (EPFL) as part of the PhD program in Mechanics. Amount: ~ 75,000 USD.
2022	EPFL-ETH Zürich Summer School Grant: given by EPFL and ETH Zürich to organize an international summer course for PhD students: “Mechanics of earthquakes and aseismic slip”, held in Zürich, Switzerland, July 2022. Amount: ~ 20,000 USD.

- 2011 **Full Tuition Fee Waiver Scholarship:** given by the Department of Civil Engineering, University of Chile, once per year to one outstanding student to study the master's program in civil engineering. Amount: $\sim 4,000$ USD.
- 2005-2010 **Bicentenario Scholarship:** given by the Chilean government to students with financial difficulties and outstanding results in the admission exam to public universities. Amount: $\sim 5,000$ USD.

PUBLICATIONS

Journal Articles - published

1. **Sáez, A.**, Lecampion, B., Bhattacharya, P. and Viesca, R.C. (2022). Three-dimensional fluid-driven stable frictional ruptures. *Journal of the Mechanics and Physics of Solids*, 132, 103672. [link]
2. **Sáez, A.** and Lecampion, B. (2023). Post-injection aseismic slip as a mechanism for the delayed triggering of seismicity. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 479, 20220810. [link]
3. **Sáez, A.** and Lecampion, B. (2024). Fluid-driven slow slip and earthquake nucleation on a slip-weakening circular fault. *Journal of the Mechanics and Physics of Solids*, 183, 105506. [link]
4. **Sáez, A.**, Passelègue, F. and Lecampion, B. (2025). Maximum size and magnitude of injection-induced slow slip events. *Science Advances*, 11, eadq0662. [link]
5. Lecampion, B., Brisson, S., Sarma, A., Gupta, A., **Sáez, A.**, Fakhretdinova, R. (2026). Three dimensional simulation of fluid-driven frictional and tensile ruptures on existing discontinuities. *Computer Methods in Applied Mechanics and Engineering*, 463, 119348. [link]

Journal Articles under review

6. Rodriguez-Padilla, A., **Sáez, A.**, and Avouac, J.-P. (2026). Earthquake sequence synchronization and alternation driven by quasistatic elastic interactions across faults. *Under review in Journal of Geophysical Research: Solid Earth [pre-print]*
7. Shi, Q., **Sáez, A.**, Li, L., Im, K., Bourne, S., van Elk, J., and Avouac, J.-P. (2026). Conditions for run-away ruptures during reservoir operations, implication for the magnitude-frequency distribution of induced earthquakes and application to Groningen. *Under review in Proceedings of the National Academy of Sciences of the United States of America. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
8. Hernández-Montenegro, J., **Sáez, A.**, Dong, J., and Asimow, P. (2026). Recent magmatism sustains liquid water in a rapidly sealing Martian crust. *Under review in Nature. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
9. Marty, S., Avouac, J.-P., Curry, W., **Sáez, A.**, da Silva, J. (2026). Earthquakes induced by hydraulic fracturing, a semi-analytical model with application to the Permian Basin. *Under review in Journal of Geophysical Research: Solid Earth.*

Journal Articles - in preparation

9. **Sáez, A.**, Kato, A., and Garagash, D. (2026). Slow slip events driven by episodic fluid release from slab dehydration. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]

10. **Sáez, A.**, and Avouac, J.-P. (2026a). Crack models for earthquake barriers. *In prep. [presented as part of the invited talk: Towards a mechanistic understanding of fault rupture segmentation at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [\[link\]](#)
11. **Sáez, A.**, and Avouac, J.-P. (2026b). Fault segmentation can be an illusion. *In prep. [presented as part of the invited talk: Towards a mechanistic understanding of fault rupture segmentation at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [\[link\]](#)
12. **Sáez, A.**, Lapusta, N., and Avouac, J.-P. (2026). Leak-off effects on fault-zone fluid transport and injection-induced aseismic slip. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [\[link\]](#)
13. **Sáez, A.**, Sarma, A., Avouac, J.-P., and Lecampion, B. (2026). Spatiotemporal patterns of earthquake swarms driven by fluid injections and induced aseismic slip. *In prep. [preliminary results presented at EGU General Assembly, Vienna, Austria, May 23-27, 2022]* [\[link\]](#)
14. Gupta, A., **Sáez, A.**, and Lecampion, B. (2026). Shearing and opening of a fault due to fluid injection. *In prep. [presented at the 26th International Congress of Theoretical and Applied Mechanics, Daegu, Korea, August 25-30, 2024]*
15. Fakhretdinova, R., **Sáez, A.**, Alcolea, A., and Lecampion, B. (2026). Revisiting the Basel-1 hydraulic stimulation with a 3D coupled hydro-mechanical model. *In prep. [preliminary results presented at 50th Stanford Geothermal Workshop, CA, USA, February 10-12, 2025]* [\[link\]](#)

Proceedings/Conference Articles

16. Lecampion, B., **Sáez, A.** and Gupta, A. (2023). Shearing and opening of a pre-existing discontinuity in response to fluid injection. *57th U.S. Rock Mechanics/Geomechanics Symposium*, Atlanta, Georgia, USA, June 25-28. [\[link\]](#)
17. Fakhretdinova, R., **Sáez, A.**, Alcolea, A., and Lecampion, B. (2025). Revisiting the Basel-1 hydraulic stimulation with a 3D coupled hydro-mechanical model. *50th Stanford Geothermal Workshop, CA, USA, February 10-12.* [\[link\]](#)

PhD Thesis

18. **Sáez, A.** (2023). Three-dimensional fluid-driven frictional ruptures: theory and applications. PhD thesis, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland. [\[link\]](#)

SERVICE

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| 2026 - now | Co-convener for session: Data-Constrained Earthquake Source Modeling: Kinematic, Dynamic, and Analytical Approaches. AGU Fall Meeting 2026, San Francisco, California |
| 2026 - now | Laboratory Safety Coordinator, Rosakis Research Group, Graduate Aerospace Laboratories, Caltech: coordinate laboratory safety practices, compliance, training, and safety documentation for the laboratory. |
| 2022 - now | Peer reviewer for Journal of Geophysical Research: Solid Earth, Earth and Planetary Science Letters, Journal of Fluid Mechanics, Philosophical Transactions of the Royal Society A, Geophysical Journal International, Rock Mechanics and Rock Engineering, Communications Earth and Environment, Nature Communications. |

2025 - now	Member of the Brown Bag Seminar organizing committee at the Seismological Laboratory of Caltech.
2022	Creator and principal organizer of the international summer course for PhD students: “Mechanics of earthquakes and aseismic slip”, Zürich, Switzerland, 18-21 July 2022.
2016-2017	Member of the technical committee CL034 of the National Standardization Institute of Chile for the development of the national code NCh3411 for the seismic design of buildings with passive energy dissipation systems.
2012	Member of conference organizing committee: Behavior of Steel Structures in Seismic Areas (STESSA) 7th International Conference, Santiago, Chile.
2010	Member of conference organizing committee: Proceedings of the 10th Chilean Conference on Seismology and Earthquake Engineering, Santiago, Chile.

TECHNOLOGY TRANSFER

2019- Today	Co-development of a Python code for simulating fluid-driven tensile and shear ruptures on existing discontinuities in 3D geological media. [link]
2019- Today	Co-development of a C++ computational code for the solution of vectorial Boundary Integral Equations accelerated with hierarchical matrices. [link]
2014	Development of a proposal for the national code for the seismic design of buildings with passive energy dissipation systems. This proposal was the main output of my master’s thesis and provided the basis for the now official national code NCh3411 of the National Standardization Institute of Chile. [link]

INVITED TALKS

04/2026	SZ4D Science Meeting 2026. “Slow slip events driven by episodic fluid release from slab dehydration”, Long Beach, LA, US.
04/2026	SSA Annual Meeting 2026, Session: Earthquake swarms across the earthquake cycle: precursors, transients and after effects. “Spatio-temporal migration patterns of micro-seismicity driven by fluid injections and induced aseismic slip”, Pasadena, LA, US.
03/2026	Department of Earth Sciences, University of Southern California - Lithosphere Dynamics Seminar Series. “Slow slip events driven by episodic fluid release from slab dehydration”, Los Angeles, CA, USA.
12/2025	AGU Fall Meeting 2025, Session S23B - Kinematic, Dynamic, and Analytical Earthquake Source Modeling. “Towards a mechanistic understanding of fault rupture segmentation”, New Orleans, LA, US.
09/2025	Chilean Society of Rock Mechanics. “Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity”, Santiago, Chile.
09/2025	American Rock Mechanics Association (ARMA) - Future Leaders Webinar Series. “Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity”, online.
05/2025	École Polytechnique Fédérale de Lausanne - On the occasion of the Mechanics Day 2025 to receive the EDME award for the best PhD thesis. “Fluid-driven frictional ruptures: theory and applications to fluid-induced seismicity”, Lausanne, Switzerland.

- 05/2025 Scripps Institution of Oceanography, UC San Diego - Geophysics seminar. "Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity", Online.
- 04/2025 Seismological Laboratory, Caltech - Seismolab seminar. "Fluid-driven shear ruptures: theoretical models and their applications to fluid-induced seismicity", Los Angeles, CA, USA.
- 03/2025 Earth and Planetary Sciences, UC Santa Cruz - IGPP Seminar. "Fluid-driven shear ruptures: theoretical models and their applications to fluid-induced seismicity", Santa Cruz, CA, USA.
- 11/2024 Department of Earth Sciences, University of Southern California - Lithosphere Dynamics Seminar. "Fluid-driven frictional ruptures: theory and applications to injection-induced seismicity", Los Angeles, CA, USA.
- 03/2024 GeoREST-Workshop on induced seismicity. "Reactivation of fault slip during and after fluid injections: theoretical considerations, micro-seismicity patterns, and aseismic moment release." Palma de Mallorca, Spain.
- 02/2023 Seismology and Rock Mechanics groups, National Research Institute for Earth Science and Disaster Prevention (NIED), Japan. "Fluid-driven stable fault slip: nucleation, propagation and arrest", Tsukuba, Japan.
- 02/2023 Seismology and Physics of the Earth's interior group, Department of Geophysics, Kyoto University. "Fluid-driven stable fault slip: nucleation, propagation and arrest", Kyoto, Japan.
- 08/2022 Seismological Laboratory, Caltech - Brown Bag Seminar. "Fluid-driven stable fault slip: nucleation, propagation and arrest", Pasadena, CA, USA.
- 08/2022 Earth Resources Laboratory, MIT. "Fluid-driven stable fault slip in 3D: nucleation, propagation and arrest", Cambridge, MA, USA.
- 08/2022 Schlumberger-Doll Research. "Fluid-driven stable fault slip: nucleation, propagation and arrest", Cambridge, MA, USA.
- 05/2022 Geoazur Laboratory, Université de Côte d'Azur. "Post-injection pulse-like aseismic ruptures", Sophia Antipolis, France.
- 06/2021 Applied and Earthquake Seismology Research Group, Department of Earth Sciences, Freie Universität Berlin. "Three-dimensional aseismic ruptures driven by fluid injection", Berlin, Germany. Online.
- 05/2021 Seismology Group, Department of Geophysics, University of Chile,. "Three-dimensional aseismic ruptures driven by fluid pressure diffusion", Santiago, Chile. Online.
- 04/2021 GFZ German Research Centre for Geosciences - Section 4.1 Lithosphere Dynamics. "Three-dimensional aseismic ruptures driven by fluid injection", Potsdam, Germany. Online.

ORAL PRESENTATIONS

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| 12/2025 | AGU Fall Meeting 2025, Session S11A - Fluids, Faulting and Earthquakes: The Role of Fluids in Seismogenesis. “Fluid flow along fault zones including leak-off into host rocks: implications for injection-induced fault slip and pore pressure distribution at depth along plate-boundary faults”, New Orleans, LA, US. |
| 10/2025 | VII Colloquium on Geophysical Signatures of Earthquakes and Volcanoes. “Towards a mechanistic understanding of fault rupture segmentation”, Valparaíso, Chile. |
| 01/2025 | Slow-to-fast Earthquake Workshop in Chile. “Slow-slip events as a fluid-driven frictional rupture process”, Santiago, Chile. |
| 10/2024 | Cargèse International Workshop on Earthquakes, 4th Edition. “Maximum size and magnitude of injection-induced slow slip events”, Cargèse, Corsica, France. |
| 05/2024 | Japan Geoscience Union (JpGU) - Meeting 2024. “Slow earthquakes and pore pressure oscillations”, Makuhari Messe, Chiba Prefecture, Japan. Online. |
| 12/2023 | American Geophysical Union (AGU) - Fall Meeting 2023. “Spatio-temporal patterns of seismic swarms driven by coupled pore pressure diffusion and aseismic slip events”, San Francisco, CA, USA. Online. |
| 05/2023 | European Geosciences Union (EGU) - General Assembly 2023. “Extraction of fluids to mitigate the seismic risk associated with post-injection aseismic slip”, Vienna, Austria. |
| 12/2022 | American Geophysical Union (AGU) - Fall Meeting 2022. “Pulse-like aseismic ruptures upon the cessation of fluid injections: regimes of propagation and arrest in 3D”, Chicago, IL, USA. Online. |
| 07/2022 | 11th European Solid Mechanics Conference (ESMC). “Three-dimensional fluid-driven stable frictional ruptures”, Galway, Ireland. |
| 05/2022 | European Geosciences Union (EGU) - General Assembly 2022. “Spatio-temporal patterns of fluid-driven aseismic slip transients: implications for seismic swarms”, Vienna, Austria. |
| 05/2022 | European Geosciences Union (EGU) - General Assembly 2022. “2D and 3D numerical modeling of fluid-driven frictional crack growth for geothermal hydraulic stimulation”, Vienna, Austria. |
| 05/2021 | European Geosciences Union (EGU) - General Assembly 2021. “Three-dimensional aseismic ruptures driven by fluid injection”, Vienna, Austria. Online. |

PROGRAMMING LANGUAGES

Proficient in C++, Python, Wolfram Mathematica, and Matlab.

LANGUAGES

Fluent in English, Spanish (native), and Portuguese.

PERSONAL DEVELOPMENT

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| 2017-2018 | Trip around the world (Australia, China, Brazil, Myanmar, among many other countries). |
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REFERENCES

Brice Lecampion

Associate Professor, Head of Geo-Energy Laboratory,
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
E-mail: brice.lecampion@epfl.ch

Jean-Philippe Avouac

Earle C. Anthony Professor of Geology and Professor of Mechanical and Civil Engineering,
California Institute of Technology.
E-mail: avouac@caltech.edu

Ares Rosakis

Theodore von Kármán Professor of Aeronautics and Professor of Mechanical Engineering,
California Institute of Technology.
Email: arosakis@caltech.edu

Eric M. Dunham

Professor of Geophysics,
Stanford University.
Email: edunham@stanford.edu

Nadia Lapusta

Lawrence A. Hanson, Jr., Professor of Mechanical Engineering and Geophysics,
California Institute of Technology.
Email: lapusta@caltech.edu

Aitaro Kato

Professor, Earthquake Research Institute, The University of Tokyo.
Email: akato@eri.u-tokyo.ac.jp

Dmitry Garagash

Professor of Civil and Resource Engineering,
Dalhousie University.
Email: garagash@dal.ca